

GLY Conjecture on Counting number of integral points in a general n -dimensional tetrahedra

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Abstract

Recently there has been tremendous interest in counting the number of integral points in n -dimensional tetrahedra with non-integral vertex due to its applications in primality testing and factoring in number theory and in singularities theory. The purpose of this paper is to summarize our previous works. A conjecture on sharp upper estimate of the number of integral points in n -dimensional tetrahedra with non-integral vertex which is called GLY conjecture. We show that this

conjecture is true for low dimensional cases as well as in the case of homogeneous n -dimensional tetrahedra.

1 Introduction

The general problem of counting the number $Q_n = Q(a_1, \dots, a_n)$ of nonnegative integral points satisfying

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$$\frac{x_1}{a_1} + \frac{x_2}{a_2} + \cdots + \frac{x_n}{a_n} \leq 1 \quad (1.1)$$

where $a_1, a_2, \dots, a_n$ are positive integers, has been a challenging problem for many years. In 1951, Mordell [Mo] gave a formula for Q_3 , expressed in terms of three Dedekind sums, in the case that a_1, a_2, a_3 are pairwise relatively prime. In 1993, Pommersheim [Po], using toric varieties, gave a formula for Q_3 for arbitrary positive integers a_1, a_2 and a_3 . More generally, let Δ be a polytope of dimension n in the lattice $\mathbf{Z}^n$. Denote $\ell_\Delta(k)$ the number of lattice points in Δ dilated by a factor k , where k is an integer:

$$\ell_\Delta(k) = \#(k\Delta \cap \mathbf{Z}^n), \quad k \in \mathbf{Z}_+ \quad (1.2)$$

Ehrhart proved that $\ell_\Delta(k)$ is a polynomial in k of degree n .

$$\ell_\Delta(k) = b_n k^n + b_{n-1} k^{n-1} + \cdots + b_0 \quad (1.3)$$

where b_n = volume of Δ , b_{n-1} = half the sum of the volumes of $(n-1)$ -dimensional faces of Δ (measured with respect to the $(n-1)$ -dimensional lattice in the $(n-1)$ -plane), $b_0=1$. In 1993, Kantor and Khovanskii [Ka-Kh] succeeded in computing b_{n-2} . In fact they gave a general formula for the number of integral points in any integral polytope in $\mathbf{R}^4$. In 1994 Cappell and Shaneson [Ca-Sh] have announced a fantastic result with which they can compute all the coefficients b_i in (1.3) explicitly. Unfortunately, in all the above mentioned results, they need to assume the vertex points of Δ are in $\mathbf{Z}^n$.

However, for the sake of applications in number theory and geometry, we are interested in the problem of estimating the number $P_n = P(a_1, \dots, a_n)$ of positive integral points satisfying (1.1), where $a_1, a_2, \dots, a_n$ are positive real numbers. Of course one can deduce an estimate for Q_n once an estimate for P_n is known and vice versa [Li-Ya 2]. In fact, let $a = \frac{1}{a_1} + \cdots + \frac{1}{a_n}$. It is easy to show that

$$P(a_1, \dots, a_n) = Q(a_1(1-a), \dots, a_n(1-a)) = Q(b_1, \dots, b_n). \quad (1.4)$$

The novelty in our problem is that we count the lattice points in a polytope whose vertices are not necessarily integer points (or even rational points). There are at least two reasons to consider this problem.

Firstly, we are told by Professor Granville [Gr] that this is an extremely important question in number theory; it would have many applications to current problems in analytic number theory, primality testing and in factoring. Given a set $\mathcal{P}$ of primes $p_1 < p_2 < \cdots < p_n \leq y$. Number theorists are interested in counting the number of integers $m \leq x$ where $m = p_1^{\ell_1} p_2^{\ell_2} \cdots p_n^{\ell_n}$ is composed only of primes from $\mathcal{P}$, and $x = y^u$, for u not too large (for all $u > 2$ would be nice, but for all $u > \log y$ would still be interesting). Thus they wish to count the number of $(\ell_1, \dots, \ell_n) \in (\mathbf{Z}_+ \cup \{0\})^n$ such that

$$\frac{\ell_1}{a_1} + \frac{\ell_2}{a_2} + \cdots + \frac{\ell_n}{a_n} \leq 1 \quad \text{where} \quad a_i = \frac{\log x}{\log p_i} \geq u$$

Observe that a_i 's are real numbers. This is, of course, the problem Q_n that we consider above. Perhaps the best reference for the application is Carl Pomerance's ICM 1994 lecture at Zürich [Pom 1], and his lecture notes [Pom 2].

Secondly, the problem has an interesting application in geometry and singularity theory. Let $f : (\mathbf{C}^n, 0) \rightarrow (\mathbf{C}, 0)$ be the germ of a complex analytic function with isolated critical point at the origin. Let M be a resolution of $V = \{(z_1, \dots, z_n) \in \mathbf{C}^n : f(z_1, \dots, z_n) = 0\}$. The Milnor number of the singularity $(V, 0)$ is $\mu = \dim \mathbf{C}\{z_1, z_2, \dots, z_n\} / (f_{z_1}, f_{z_2}, \dots, f_{z_n})$. The geometric genus p_g of the singularity $(V, 0)$ is $\dim H^{n-2}(M, \mathcal{O})$, which is an important invariant of the singularity. In 1978 Durfee [Du] made the following conjecture.

Durfee Conjecture $n! p_g \leq \mu$ with equality only when $\mu = 0$.

The connection between Durfee conjecture and the proposed problem is as follows. A polynomial $f(z_1, \dots, z_n)$ is weighted homogeneous of type $(w_1, \dots, w_n)$, where $w_1, \dots, w_n$ are fixed positive rational numbers, if f can be expressed as a linear combination of monomials $z_1^{i_1} \dots z_n^{i_n}$ for which $i_1/w_1 + \dots + i_n/w_n = 1$. If $f(z_1, \dots, z_n)$ is a weighted homogeneous polynomial of type $(a_1, \dots, a_n)$ with an isolated singularity at the origin, then Milnor and Orlik [Mi-Or] proved that $\mu = (a_1 - 1)(a_2 - 1) \dots (a_n - 1)$. On the other hand, Merle and Teissier [Me-Te] showed that p_g is exactly the number P_n of positive integral points satisfying (1.1). Thus Durfee conjecture provides us a guidance for the upper estimate of P_n . Unfortunately Durfee conjecture is not sharp, see for example [Xu-Ya 2]. In 1995, the second author has made the following conjecture.

Conjecture Let $f : (\mathbf{C}^n, 0) \rightarrow (\mathbf{C}, 0)$ be a weighted homogeneous polynomial with an isolated critical point at the origin. Then

$$\mu - h(\nu) \geq n! p_g$$

and equality holds if and only if f is a homogeneous polynomial. Here $h(\nu)$ is a polynomial function on the multiplicity ν with the properties $h(\nu) \geq 0$ and $h(\nu) = 0$ if and only if $\nu = 1$.

Counting the number of integral points in an n -dimensional tetrahedra with non-integral vertices is a classical subject which has attracted a lot of famous mathematicians. For $n = 2$, Hardy and Littlewood [Ha-Li 2] [Ha-Li 3] wrote two famous papers on the lattice points of a right-angled triangle because of its relations to their International Congress of Mathematics lecture [Ha-Li 1] in 1912 on Diophantine approximation. A general result was obtained by D.C. Spencer [Sp 1] [Sp 2] in 1942 via complex function-theoretic methods, as follows. Write $a_i = x/\rho_i$, $1 \leq i \leq n$. Then

$$Q(a_1, \dots, a_n) = \frac{1}{n!} a_1 \dots a_n + \frac{1}{2(n-1)!} \left(\frac{a_n}{a_1} + \dots + \frac{a_n}{a_{n-1}} + 1 \right) (a_1 \dots a_{n-1}) + o(x^{n-1}).$$

In 1975, Beukers [Be] used an elementary method to reprove the above formula. Unlike the

sharp upper estimate in our Main Theorem, the above formula of Hardy-Littlewood-Spencer is only asymptotic.

Rosser [Ro] in 1939 obtained a lower bound for $Q(a_1, \dots, a_n)$ as a polynomial $R(a_1, \dots, a_n)$ which, when extended to the general tetrahedron, may be written

$$R(a_1, \dots, a_n) = \frac{1}{n! \rho_1 \rho_2 \dots \rho_n} \left(x^n + \frac{n}{2} \sigma'_1 x^{n-1} + \frac{n(n-1)}{2^2} \sigma'_2 x^{n-2} + \dots + \frac{n!}{2^{n-1}} \sigma'_{n-1} x \right)$$

where σ'_k is the sum of the products k at a time of $\rho_2, \rho_3, \dots, \rho_n$. Lehmer [Le] constructed two polynomials $\ell(a_1, \dots, a_n)$ and $L(a_1, \dots, a_n)$ which approximate $Q(a_1, \dots, a_n)$ from below and above respectively. $\ell(a_1, \dots, a_n)$ and $L(a_1, \dots, a_n)$ are polynomials of degree n in x with coefficients depending on $\rho_1, \dots, \rho_n$. The first two coefficients of $\ell(a_1, \dots, a_n)$ are seen to agree with those of $R(a_1, \dots, a_n)$. For $n > 2$ and x large enough, Lehmer asserted that

$$R(a_1, \dots, a_n) < \ell(a_1, \dots, a_n) < Q(a_1, \dots, a_n).$$

Let $(\rho_1, \dots, \rho_5)$ be $(\log_{10} 2, \log_{10} 3, \log_{10} 5, \log_{10} 5, \log_{10} 7, \log_{10} 11)$. A.E. Western has prepared extensive tables of $Q(a_1, a_2, a_3, a_4, a_5)$. From these tables, he has constructed an approximating polynomial $w(a_1, a_2, \dots, a_n)$ by applying the method of least squares. Unfortunately, all these polynomials are far from being sharp. In particular, they are not useful for the geometric applications mentioned above.

If $f(z)$ is a weighted homogeneous polynomial of type $(w_1, \dots, w_n)$, then the Milnor number μ is given by $\mu = (w_1 - 1)(w_2 - 1) \dots (w_n - 1)$. Therefore Durfee Conjecture is a special case of the following conjecture.

Conjecture 1.1 *Let $a_1, \dots, a_n$ be positive real numbers greater than or equal to two. Then*

$$n! P_{(a_1, \dots, a_n)} \leq (a_1 - 1)(a_2 - 1) \dots (a_n - 1). \quad (1.5)$$

The estimate in the above conjecture is sharper than the following polynomial estimate (1.7) provided by number theorist. Attach a unit cube to the right and above each lattice point of $\Delta(a_1, \dots, a_n)$. Then

$$\begin{aligned} Q_{(a_1, \dots, a_n)} &= \Sigma \text{ volume of the unit cube attached to each lattice point} \\ &\leq \text{volume of } \left\{ (x_1, \dots, x_n) \in \mathbf{R}_+^n : \sum_{i=1}^n \frac{x_i - 1}{a_i} \leq 1 \right\} \\ &= \frac{1}{n!} \left(\prod_{i=1}^n a_i \right) \left(1 + \sum_{j=1}^n \frac{1}{a_j} \right)^n \end{aligned} \quad (1.6)$$

Hence by (1.4) and (1.5), we have

$$\begin{aligned} P_{(a_1, \dots, a_n)} &= Q_{(b_1, \dots, b_n)} \\ &\leq \frac{1}{n!} \left(\prod_{i=1}^n b_i \right) \left(1 + \sum_{j=1}^n \frac{1}{b_j} \right)^n \\ &= \frac{1}{n!} \prod_{i=1}^n a_i. \end{aligned} \quad (1.7)$$

The estimate of $P_{(a_1, \dots, a_n)}$ given by (1.5) is nice. However it is not sharp enough to provide a solution of the following thirty year old problem in singularities theory.

Problem. Let $f : (\mathbb{C}^n, 0) \longrightarrow (\mathbb{C}, 0)$ be a complex analytic function with isolated critical point at the origin. Find an intrinsic characterization for f to be a homogeneous polynomial.

In order to solve the above problem, we need a sharp upper polynomial estimate of $P_{(a_1, \dots, a_n)}$ such that equality holds if and only if $a_1 = a_2 = \dots = a_n = \text{integer}$. The intrinsic characterization of homogeneous polynomial problem was solved for $n = 3$ by Xu and Yau [Xu-Ya1] and for $n = 4$ by Lin and Yau [Li-Ya1]. The purpose of this paper is to formulate a conjectural sharp polynomial upper estimate for $P_{(a_1, \dots, a_n)}$ such that equality holds if and only if $a_1 = a_2 = \dots = a_n = \text{integer}$. In section 2, we show that our conjecture is true for $n = 3, 4$ and 5. In section 3, we give an exact formula for $P_{(a_1, \dots, a_n)}$ for homogeneous n -dimensional tetrahedra (i.e. $a_1 = \dots = a_n$). We show that our conjecture is also true in this case.

2 Sharp polynomial upper estimate

Before we formulate our sharp polynomial upper estimate conjecture, it is convenient to introduce some notations. Let $a, a_1, \dots, a_n$ be positive real numbers greater than or equal to $n - 1$. We shall denote

$$S_k^{n-1} = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n-1} i_1 i_2 \dots i_k, \quad S_0^{n-1} = 1, \quad S_{n-1}^{n-1} = 1 \cdot 2 \dots (n-1) \quad (2.1)$$

where $i_1, i_2, \dots, i_k$ are integers. Then

$$\begin{aligned} a(a-1)(a-2) \dots (a-(n-1)) &= a^n - \left(\sum_{i_1=1}^{n-1} i_1 \right) a^{n-1} + (-1)^2 \sum_{1 \leq i_1 < i_2 \leq n-1} i_1 i_2 a^{n-2} \\ &\quad + \dots + (-1)^k \sum_{1 \leq i_1 < \dots < i_k \leq n-1} i_1 i_2 \dots i_k a^{n-k} + \dots + (-1)^{n-1} \left(\prod_{i=1}^{n-1} i \right) a \\ &= a^n + (-1) S_1^{n-1} a^{n-1} + (-1)^2 S_2^{n-1} a^{n-2} + \dots + (-1)^k S_k^{n-1} a^{n-k} + \dots + (-1)^{n-1} S_{n-1}^{n-1} a. \end{aligned} \quad (2.2)$$

We shall denote

$$A_{n-k}^n = \left(\prod_{i=1}^n a_i \right) \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \frac{1}{a_{i_1} a_{i_2} \dots a_{i_k}} \quad (2.3)$$

$$A_n^n = \prod_{i=1}^n a_i, \quad A_0^n = 1. \quad (2.4)$$

Observe that A_{n-k}^n is a polynomial in $a_1, \dots, a_n$ of degree $n - k$.

The following is our conjecture on sharp polynomial estimate for $P_{(a_1, \dots, a_n)}$.

Main Conjecture. Let $P_{(a_1, \dots, a_n)} = \# \left\{ (x_1, \dots, x_n) \in \mathbb{Z}_+^n : \frac{x_1}{a_1} + \dots + \frac{x_n}{a_n} \leq 1 \right\}$, where $a_1 \geq a_2 \geq \dots \geq a_n \geq n-1$, $n \geq 3$. Then

$$\begin{aligned}
n!P_{(a_1, \dots, a_n)} &\leq \prod_{i=1}^n a_i + (-1) \frac{S_1^{n-1}}{n} \left(\prod_{i=1}^n a_i \right) \left(\sum_{k=1}^n \frac{1}{a_k} \right) + (-1)^2 \frac{S_2^{n-1}}{\binom{n-1}{1}} \left(\prod_{i=1}^{n-1} a_i \right) \left(\sum_{k=1}^{n-1} \frac{1}{a_k} \right) \\
&+ (-1)^3 \frac{S_3^{n-1}}{\binom{n-1}{2}} \left(\prod_{i=1}^{n-1} a_i \right) \left(\sum_{1 \leq i_1 < i_2 \leq n-1} \frac{1}{a_{i_1} a_{i_2}} \right) \\
&+ (-1)^4 \frac{S_4^{n-1}}{\binom{n-1}{3}} \left(\prod_{i=1}^{n-1} a_i \right) \left(\sum_{1 \leq i_1 < i_2 < i_3 \leq n-1} \frac{1}{a_{i_1} a_{i_2} a_{i_3}} \right) \\
&+ \dots + \\
&+ (-1)^{k+1} \frac{S_{k+1}^{n-1}}{\binom{n-1}{k}} \left(\prod_{i=1}^{n-1} a_i \right) \left(\sum_{1 \leq i_1 < \dots < i_k \leq n-1} \frac{1}{a_{i_1} a_{i_2} \dots a_{i_k}} \right) \\
&+ \dots + \\
&+ (-1)^{n-1} \frac{S_{n-1}^{n-1}}{\binom{n-1}{n-2}} (a_1 + a_2 + \dots + a_{n-1}) \\
&= A_n^n + (-1) \frac{S_1^{n-1}}{n} A_{n-1}^n + (-1)^2 \frac{S_2^{n-1}}{\binom{n-1}{1}} A_{n-2}^{n-1} + (-1)^3 \frac{S_3^{n-1}}{\binom{n-1}{2}} A_{n-3}^{n-1} \\
&+ (-1)^4 \frac{S_4^{n-1}}{\binom{n-1}{3}} A_{n-4}^{n-1} + \dots + (-1)^{k+1} \frac{S_{k+1}^{n-1}}{\binom{n-1}{k}} A_{n-k-1}^{n-1} + \dots + (-1)^{n-1} \frac{S_{n-1}^{n-1}}{\binom{n-1}{n-2}} A_1^{n-1} \quad (2.5)
\end{aligned}$$

and the equality holds if and only if $a_1 = a_2 = \dots = a_n = \text{integer}$.

For $n = 3$, the Main Conjecture asserts that for $a_1 \geq a_2 \geq a_3 \geq 2$,

$$\begin{aligned}
3!P_{(a_1, a_2, a_3)} &\leq A_3^3 - \frac{S_1^2}{3} A_2^3 + \frac{S_2^2}{\binom{2}{1}} A_1^2 \\
&= a_1 a_2 a_3 - \frac{1+2}{3} a_1 a_2 a_3 \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} \right) + \frac{2}{2} a_1 a_2 \left(\frac{1}{a_1} + \frac{1}{a_2} \right) \\
&= a_1 a_2 a_3 - (a_1 a_2 + a_1 a_3 + a_2 a_3) + (a_1 + a_2)
\end{aligned}$$

with equality if and only if $a_1 = a_2 = a_3 = \text{integer}$. This is the main result proved by Xu and Yau in their paper [Xu-Ya1].

For $n = 4$, the Main Conjecture asserts that for $a_1 \geq a_2 \geq a_3 \geq a_4 \geq 3$,

$$4!P_{(a_1, a_2, a_3, a_4)} \leq A_4^4 + (-1) \frac{S_1^3}{4} A_3^4 + (-1)^2 \frac{S_2^3}{\binom{3}{1}} A_2^3 + (-1)^3 \frac{S_3^3}{\binom{3}{2}} A_1^3$$

$$\begin{aligned}
&= a_1 a_2 a_3 a_4 - \frac{1+2+3}{4} a_1 a_2 a_3 a_4 \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} \right) \\
&\quad + \frac{1 \cdot 2 + 1 \cdot 3 + 2 \cdot 3}{3} a_1 a_2 a_3 \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} \right) \\
&\quad - \frac{1 \cdot 2 \cdot 3}{3} a_1 a_2 a_3 \left(\frac{1}{a_1 a_2} + \frac{1}{a_1 a_3} + \frac{1}{a_2 a_3} \right) \\
&= a_1 a_2 a_3 a_4 - \frac{3}{2} (a_1 a_2 a_3 + a_1 a_2 a_4 + a_1 a_3 a_4 + a_2 a_3 a_4) \\
&\quad + \frac{11}{3} (a_1 a_2 + a_1 a_3 + a_2 a_3) - 2(a_1 + a_2 + a_3)
\end{aligned}$$

with equality if and only if $a_1 = a_2 = a_3 = a_4 = \text{integer}$. Xu and Yau [Xu-Ya3] proved that if $a_1 \geq a_2 \geq a_3 \geq a_4 \geq 2$ and $P_{(a_1, a_2, a_3, a_4)} > 0$, then the above result is true. Lin and Yau [Li-Ya1] proved that the condition $P_{(a_1, a_2, a_3, a_4)} > 0$ can be removed if we assume $a_1 \geq a_2 \geq a_3 \geq a_4 \geq 3$. Therefore our Main Conjecture is true for $n = 4$.

For $n = 5$, the Main Conjecture asserts that for $a_1 \geq a_2 \geq a_3 \geq a_4 \geq a_5 \geq 4$.

$$\begin{aligned}
&5!P_{(a_1, a_2, a_3, a_4, a_5)} \leq A_5^5 + (-1) \frac{S_1^4}{5} A_4^5 + \frac{S_2^4}{\binom{4}{1}} A_3^4 - \frac{S_3^4}{\binom{4}{2}} A_2^4 + \frac{S_4^4}{\binom{4}{3}} A_1^4 \\
&= a_1 a_2 a_3 a_4 a_5 - \frac{1+2+3+4}{5} a_1 a_2 a_3 a_4 a_5 \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} + \frac{1}{a_5} \right) \\
&\quad + \frac{(1 \cdot 2 + 1 \cdot 3 + 1 \cdot 4 + 2 \cdot 3 + 2 \cdot 4 + 3 \cdot 4)}{4} a_1 a_2 a_3 a_4 \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} \right) \\
&\quad - \frac{(1 \cdot 2 \cdot 3 + 1 \cdot 2 \cdot 4 + 1 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4)}{\binom{4}{2}} a_1 a_2 a_3 a_4 \\
&\quad \quad \left(\frac{1}{a_1 a_2} + \frac{1}{a_1 a_3} + \frac{1}{a_1 a_4} + \frac{1}{a_2 a_3} + \frac{1}{a_2 a_4} + \frac{1}{a_3 a_4} \right) \\
&\quad + \frac{1 \cdot 2 \cdot 3 \cdot 4}{4} a_1 a_2 a_3 a_4 \left(\frac{1}{a_1 a_2 a_3} + \frac{1}{a_1 a_2 a_4} + \frac{1}{a_2 a_3 a_4} \right) \\
&= a_1 a_2 a_3 a_4 a_5 - 2(a_1 a_2 a_3 a_4 + a_1 a_2 a_3 a_5 + a_1 a_2 a_4 a_5 + a_1 a_3 a_4 a_5 + a_2 a_3 a_4 a_5) \\
&\quad + \frac{35}{4} (a_1 a_2 a_3 + a_1 a_2 a_4 + a_1 a_3 a_4 + a_2 a_3 a_4) \\
&\quad - \frac{25}{3} (a_1 a_2 + a_1 a_3 + a_1 a_4 + a_2 a_3 + a_2 a_4 + a_3 a_4) \\
&\quad + 6(a_1 + a_2 + a_3 + a_4)
\end{aligned}$$

with equality if and only if $a_1 = a_2 = a_3 = a_4 = a_5 = \text{integer}$. This is the main result proved by Lin and Yau in their paper [Li-Ya3].

3 Number of positive integral points in a homogeneous n -dimensional tetrahedra

In this section we shall prove that our Main Conjecture in section 2 holds for homogeneous n -dimensional tetrahedra $\Delta(a, a, \dots, a)$. We first start with an elementary summation by parts lemma.

Lemma 3.1 *Let $\{u_k\}, \{v_k\}, k = 1, \dots, n$ be two sequence of numbers. Then*

$$\sum_{k=1}^n u_k v_k = u_n \sum_{k=1}^n v_k - \sum_{r=2}^n \left[(u_r - u_{r-1}) \sum_{k=1}^{r-1} v_k \right]. \quad (3.1)$$

PROOF: We shall prove (3.1) by induction. (3.1) is obviously true for $n = 2$. Assume that (3.1) holds for $n - 1$. Then

$$\begin{aligned} \sum_{k=1}^n u_k v_k &= \sum_{k=1}^{n-1} u_k v_k + u_n v_n \\ &= u_{n-1} \sum_{k=1}^{n-1} v_k - \sum_{r=2}^{n-1} \left[(u_r - u_{r-1}) \sum_{k=1}^{r-1} v_k \right] + u_n v_n \\ &= u_n \sum_{k=1}^n v_k - u_n \sum_{k=1}^{n-1} v_k + u_{n-1} \sum_{k=1}^{n-1} v_k - \sum_{r=2}^{n-1} \left[(u_r - u_{r-1}) \sum_{k=1}^{r-1} v_k \right] \\ &= u_n \sum_{k=1}^n v_k - (u_n - u_{n-1}) \sum_{k=1}^{n-1} v_k - \sum_{r=2}^{n-1} \left[(u_r - u_{r-1}) \sum_{k=1}^{r-1} v_k \right] \\ &= u_n \sum_{k=1}^n v_k - \sum_{r=2}^n \left[(u_r - u_{r-1}) \sum_{k=1}^{r-1} v_k \right] \end{aligned}$$

Q.E.D.

Theorem 3.1 *Let $n \geq 2$ be a positive integer and $a \geq n$. Let P_n be the number of positive integral solution of $\frac{x_1}{a} + \frac{x_2}{a} + \dots + \frac{x_n}{a} \leq 1$, i.e. $P_n = \#\{(x_1, x_2, \dots, x_n) \in \mathbb{Z}_+^n : \frac{x_1}{a} + \frac{x_2}{a} + \dots + \frac{x_n}{a} \leq 1\}$. Let $[a]$ be the greatest integer less than or equal to a . Then*

$$\begin{aligned} n!P_n &= [a]([a] - 1)([a] - 2) \cdots ([a] - n + 1) \\ &\leq a(a - 1)(a - 2) \cdots (a - n + 1) \end{aligned} \quad (3.2)$$

with equality if and only if a is an integer.

PROOF: For $n = 2$, we have $\frac{x_1}{a} + \frac{x_2}{a} \leq 1$. From the level $x_2 = 1$ to the level $x_2 = [a] - 1$, we have the following positive integral points:

$$\begin{aligned} x_2 = 1 & : (1, 1), (2, 1), \dots, ([a] - 1, 1) \\ x_2 = 2 & : (1, 2), (2, 2), \dots, ([a] - 2, 2) \\ & \vdots \\ x_2 = [a] - 1 & : (1, [a] - 1). \end{aligned}$$

Hence $P_2 = 1 + 2 + \dots + ([a] - 1) = \frac{[a]([a]-1)}{2}$, i.e. $2!P_2 = [a]([a] - 1)$.

Now assume that (3.2) holds for any integer n . Consider a homogeneous $(n+1)$ -dimensional tetrahedra $\frac{x_1}{a} + \frac{x_2}{a} + \dots + \frac{x_{n+1}}{a} \leq 1$.

From the level $x_{n+1} = 1$ to the level $x_{n+1} = [a] - n$, we have the following homogeneous n -dimensional tetrahedra:

$$\begin{aligned} x_{n+1} = 1 & : \frac{x_1}{a-1} + \frac{x_2}{a-1} + \dots + \frac{x_n}{a-1} \leq 1 \\ x_{n+1} = 2 & : \frac{x_1}{a-2} + \frac{x_2}{a-2} + \dots + \frac{x_n}{a-2} \leq 1 \\ & \vdots \\ x_{n+1} = [a] - n & : \frac{x_1}{a-[a]+n} + \frac{x_2}{a-[a]+n} + \dots + \frac{x_n}{a-[a]+n} \leq 1. \end{aligned}$$

Hence by induction hypothesis we have

$$\begin{aligned} n!P_{n+1} &= \sum_{k=n}^{[a]-1} k(k-1)\dots(k-(n-1)) \\ &= \sum_{k=1}^{[a]-1} k(k-1)\dots(k-(n-1)). \end{aligned}$$

Now we apply Lemma 3.1 with $u_k = (k-1)(k-2)\dots(k-(n-1))$, $v_k = k$. Then $\sum_{k=1}^{r-1} v_k =$

$$\sum_{k=1}^{r-1} k = \frac{r(r-1)}{2}$$

$$\begin{aligned} u_r - u_{r-1} &= (r-1)(r-2)\dots(r-(n-1)) - (r-2)(r-3)\dots(r-n) \\ &= (r-2)(r-3)\dots(r-n+1)((r-1) - (r-n)) \\ &= (n-1)(r-2)(r-3)\dots(r-n+1). \end{aligned}$$

From (3.1), we have

$$\sum_{k=1}^{[a]-1} k(k-1)\dots(k-n+1)$$

$$\begin{aligned}
&= u_{[a]-1} \frac{[a]([a]-1)}{2} - \sum_{r=2}^{[a]-1} (n-1)(r-2)(r-3) \dots (r-n+1) \frac{r(r-1)}{2} \\
&= ([a]-2)([a]-3) \dots ([a]-n) \frac{[a]([a]-1)}{2} - \frac{n-1}{2} \sum_{k=1}^{[a]-1} k(k-1)(k-2) \dots (k-n+1).
\end{aligned}$$

Hence we have

$$\begin{aligned}
&\left(1 + \frac{n-1}{2}\right) \sum_{k=1}^{[a]-1} k(k-1)(k-2) \dots (k-n+1) \\
&= \frac{1}{2} [a]([a]-1)([a]-2) \dots ([a]-n),
\end{aligned}$$

which implies

$$(n+1) \sum_{k=1}^{[a]-1} k(k-1)(k-2) \dots (k-n+1) = [a]([a]-1)([a]-2) \dots ([a]-n).$$

Therefore we have

$$\begin{aligned}
n!P_{n+1} &= \sum_{k=1}^{[a]-1} k(k-1)(k-2) \dots (k-n+1) \\
&= \frac{1}{n+1} [a]([a]-1)([a]-2) \dots ([a]-n)
\end{aligned}$$

i.e. $(n+1)!P_{n+1} = [a]([a]-1)([a]-2) \dots ([a]-n)$. *Q.E.D.*

Corollary 3.1 *The Main Conjecture in section 2 is true for n -dimensional homogeneous tetrahedra $\Delta(a, a, \dots, a)$.*

PROOF: Set $a_1 = a_2 = \dots = a_n = a$ in the Main Conjecture. We get

$$\begin{aligned}
n!P_{(a, \dots, a)} &\leq a^n + (-1)S_1^{n-1}a^{n-1} + (-1)^2S_2^{n-1}a^{n-2} + (-1)^3S_3^{n-1}a^{n-3} \\
&\quad + \dots + (-1)^{k+1}S_{k+1}^{n-1}a^{n-k-1} + \dots + (-1)^{n-1}S_{n-1}^{n-1}a \\
&= a(a-1)(a-2) \dots (a-(n-1)) \quad \text{by (2.2)}
\end{aligned}$$

This is exactly Theorem 3.1. *Q.E.D.*

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